Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

When a substance A reacts with water it produces a combustible gas B and a solution of a substance C in water. When another substance D reacts with this solution of C, it also produces the same gas B on warming, but D can produce B on reaction with dilute sulphuric acid at room temperature. A imparts a deep golden yellow colour to a smokeless flame of Bunsen burner. A, B, C and D respectively are -

(A) $Na, H_2, NaOH$ and Zn

(B) K, H_2, KOH and Al

(C) $Ca, H_2, Ca(OH)_2$ and Sn

(D) $CaC_2, C_2H_2, Ca(OH)_2$ and Sn

Q.2

A colourless solid (X) on heating evolved CO_2 and also gave a white residue, soluble in water. Residue also gave CO_2 when treated with dilute acid (X) is -

(A) Na_2CO_3

(B) $CaCO_3$

(C) $NaHCO_3$

(D) $Ca\,(HCO_3)_2$

Q.3

CO_2 gas along with solid (Y) is obtained when sodium salt (X) is heated. (X) is again obtained when CO_2 gas is passed (X) and (Y) are -

(A) Na_2CO_3, Na_2O

(B) $Na_2CO_3, NaOH$

(C) $NaHCO_3, Na_2CO_3$

(D) $Na_2CO_3,\ NaHCO_3$

Q.4

Consider the following statements ; S_1 : Among alkali metal halides lithium iodide is the most covalent in nature.
S_2 : Potassium has greater photoelectric work function than sodium.
S_3 : The blue solution of alkali metals in liquid ammonia is stable at room temperature, where ammonia is still a liquid, in the presence of Fe.

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S_4 : The melting and boiling points of alkali metal halides always follow the trend : chloride $>$ fluoride $>$ bromide $>$ iodide. and arrange in the order of true/false.

(A) T F F F

(B) T T F F

(C) T F T T

(D) T T T F

Q.5

25ml of H_2O_2 solution were added to excess of acidified KI solution. The iodine so liberated required 20ml of $0.1\ \text{N} Na_2\ S_2O_3$ solution. Calculate strength in terms of normality and percentage.

(A) $0.04\ \text{N}, 0.136\%$

(B) $0.08\ \text{N}, 0.136\%$

(C) $0.08\ \text{N}, 0.163\%$

(D) $0.02\ \text{N}, 0.163\%$

Q.6

The critical temperature of water is higher than that of O_2 because H_2O molecule has

(A) Fewer electrons than oxygen

(B) Two covalent bonds

(C) V-shape

(D) Dipole moment

Q.7

The amount of H_2O_2 present in $1\ \text{L}$ of $1.5\ \text{N}\ H_2O_2$ solution is $-$

(A) $2.5\ \text{g}$

(B) $25.5\ \text{g}$

(C) $3.0\ \text{g}$

(D) $8.0\ \text{g}$

Q.8

If, one assumes linear structure instead of bent structure for water, then which one of the following properties cannot be explained?

(A) The formation of intermolecular hydrogen bonds in water.

(B) The high boiling point of water

(C) Solubility of polar compounds in water

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(D) Ability of water to form co-ordinate covalent bonds.

Q.9

Which of the following statement(s) is/are incorrect.

(A) Alkali metals are less electropositive than alkaline earth metals.

(B) The alkaline earth metals are more denser and harder than alkali metals.

(C) The alkali metals are denser and harder than alkaline earth metals.

(D) The first IP of alkali metals is more than that of alkaline earth metals.

Q.10

Which of the following statements regarding isolation of Na and K metal is/are false -

(A) These metals can be isolated by the reduction of their oxides

(B) These metals can be prepared from their aqueous salt solutions by metal displacement

(C) These metals can be prepared by the electrolysis of their aqueous salt solutions

(D) These metals can be prepared by the electrolysis of their fused chlorides

Q.11

Choose the correct statements -

(A) Sodium oxide is more basic then magnesium oxide

(B) Beryllium oxide is amphoteric

(C) The thermal stability of beryllium carbonate is more than that of calcium carbonate

(D) Beryllium is amphoteric

Q.12

Paragraph:
It is possible to measure the level of hard water by obtaining a free water testing kit. These are supplied by most water softening companies. There are different scales used to describe the hardness of water in different contexts.
(i) $\mathrm{mmol/L}$ (millimoles per litre)
(ii) $\mathrm{mg/L}$ calcium carbonate equivalent
(iii) grains/gallon (9p9)
(iv) Gr/U.S. gal $= 17.11\mathrm{mg/L}$
(v) Parts per million weight/volume (ppm w/v or ppm $\mathrm{m/v}$) various obsolete "degree".
* Clark degrees $(^{\circ}\mathrm{Clark}/$ English degrees $(^{\circ}\mathrm{E})$, conversion to $\mathrm{mg/L}$ Calcium divide by 0.175. One degree Clark corresponds to one grain of calcium carbonate in one imperial gallon of water which is equal to 14.28 parts calcium carbonate in $1,000,000$ parts water.
* Deutsche Harte (German hardness) $(^{\circ}\mathrm{dH})$ conversion to $\mathrm{mg/L}$ calcium divide by 0.14. One degree German hardness corresponds to one part calcium oxide in 100,000 parts of water.
* French degree $(^{\circ}\mathrm{f})$ conversion to $\mathrm{mg/L}$ calcium, divide by 0.25. One degree French corresponds to one part

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calcium carbonate is 100,000 parts of water.
* One degree American corresponds to one part calcium carbonate is $1,000,000$ parts water ($1 \mathrm{mg/L}$ or $1 \mathrm{ppm}) \mathrm{dGH}$. One degree of general hardness corresponds to $10 \mathrm{mg}$ of calcium oxide or magnesium oxide per litre of water.

Question:

The hardness of a sample of water is expressed as $120 \mathrm{ppm}$. It is equivalent to

(A) $8.4^{\circ} \mathrm{E}$

(B) $7.4^{\circ} \mathrm{dH}$

(C) $24 \mathrm{mg/L}$

(D) All are correct

Q.13

Paragraph:

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Question:

Tap water has a $200 \mathrm{mg/L}$ calcium carbonate. It corresponds to

(A) 1.12

(B) 11.2

(C) 112

(D) 56

Q.14

Paragraph:

It is possible to measure the level of hard water by obtaining a free water testing kit. These are supplied by most water softening companies. There are different scales used to describe the hardness of water in different contexts.

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(i) $\mathrm{mmol/L}$ (millimoles per litre)
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* One degree American corresponds to one part calcium carbonate is $1,000,000$ parts water ($1\mathrm{mg/L}$ or $1\mathrm{ppm})\mathrm{dGH}$. One degree of general hardness corresponds to $10\mathrm{mg}$ of calcium oxide or magnesium oxide per litre of water.

Question:

$10\mathrm{dGH}$ is expressed in Americal degree of hardness as

(A) 178.6 ppm

(B) $1.786\mathrm{ppm}$

(C) $100\mathrm{ppm}$

(D) $10\mathrm{ppm}$

Q.15

Paragraph:

A compound X on heating gives a colourless gas. The residue is dissolved in water to obtain Y. Excess $\mathrm{CO_2}$ is passed through aqueous solution of Y when Z is formed. Z on gentle heating gives back X.

Question:

The compound X is -

(A) $\mathrm{Na_2CO_3}$

(B) $\mathrm{CaCO_3}$

(C) $\mathrm{Ca\,(HCO_3)_2}$

(D) $\mathrm{NaHCO_3}$

Q.16

Paragraph:

A compound X on heating gives a colourless gas. The residue is dissolved in water to obtain Y. Excess $\mathrm{CO_2}$ is passed through aqueous solution of Y when Z is formed. Z on gentle heating gives back X.

Question:

The compound Y is -

(A) $\mathrm{Na_2CO_3}$

(B) $\mathrm{Ca(OH)_2}$

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(C) $Ca(HCO_3)_2$

(D) $NaHCO_3$

Q.17

Paragraph:

A compound X on heating gives a colourless gas. The residue is dissolved in water to obtain Y. Excess CO_2 is passed through aqueous solution of Y when Z is formed. Z on gentle heating gives back X.

Question:

The compound Z is -

(A) Na_2CO_3

(B) $CaCO_3$

(C) $Ca(HCO_3)_2$

(D) $NaHCO_3$

Q.18

Match the column

Column I	**Column II**
(A) Absorbs N_2 gas	(p) Liquid lithium
(B) Coolent in heat exchangers	(q) Barium dissolved in liquid sodium
(C) Reduction of halide with molten potassium	(r) Alloy of sodium and potassium
(D) Large liquid range	(s) Hot metallic magnesium

(A) A-q,s; B-r ; C- s ; D-p;

(B) A-p,q,r,; B-r ; C- s ; D-s;

(C) A-s; B-r,p,q ; C-s ; D-s;

(D) A-s,r; B-p ; C- p ; D-s;

Q.19

How many types of product are formed when $LiNO_3$ and $NaNO_3$ are heated at $500^\circ C$.

Q.20

Number of non-radioactive alkali metals forming superoxide as major product on heating with excess $O_2 = x$,
Number of II-A metals of periodic table whose hydrated halides suffer hydrolysis on heating $= y$.
Number of non radio active alkali metals which dissolve in liquid ammonia and produce blue colour solution $= u$
The value of $(x + y - u)$ would be....

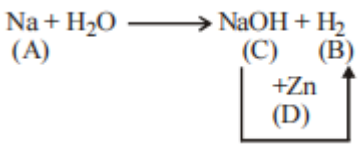
Answers & Solutions

Q.1 Answer:

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Solution:

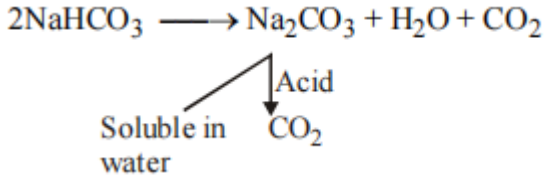


$$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$$

Q.2 Answer:

$NaHCO_3$

Solution:



Q.3 Answer:

$NaHCO_3, Na_2CO_3$

Solution:

$$\underset{(X)}{2NaHCO_3} \xrightarrow{\text{Heat}} \underset{(Y)}{Na_2CO_3} + H_2O + CO_2;$$

$$Na_2CO_3 + H_2O + CO_2 \longrightarrow 2NaHCO_3$$

Q.4 Answer:

T F F F

Solution:

S_1 : Li^+ being smaller have high polarising power and I^- being larger have high polarisability. So it is most covelent among alkali metal halides according to Fajan's rule.

S_2 : The IE1 of potussium atom is less then sodium atom

S_3 : The presence of transition metals like iron and other impurities catalyses the decompositon of deep blue solution forming amide and liberating H_2.

S_4 : Two opposing tendencies exists. With greater charge and smaller size of cation, lattice energy increases which tends to increase the melting point ; while increase in covalent character causes a decrease in melting point. Hence, no unique generalised trend may be stated for melting points. (Students need not worry about or memorise such experimental data).

Q.5 Answer:

$0.08\ N, 0.136\%$

Solution:

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$O_2^{2-} + 2e^- \rightarrow 2O^{-2}; \quad 2I^- \rightarrow I_2 + 2e^-$

$2\left(S^{+2}\right)_2 \rightarrow \left(S^{+5/2}\right)_4 + 2e^-; I_2 + 2e^- \rightarrow 2I^-$

Meq. of H_2O_2 = **Meq. of** I_2 = **Meq. of** $Na_2\ S_2O_3$

$$\frac{W}{34/2} \times 1000 = 20 \times 0.1$$

$$\therefore W_{H_2O_2} = 0.034 gr/25 ml$$

$$\therefore N_{H_2O_2} = \frac{0.034}{34/2} \times \frac{1000}{25} = 0.08$$

Vol. strength $= 5.6 \times 0.08 = 0.448$

% $\frac{17}{56} \times 5.6 \times 0.08 = 0.136\%$

Q.6 Answer:

Dipole moment

Solution:

Critical temperature of water is more than O_2 **due to its dipole moment (Dipole moment of water** $= 1.84D$; **Dipole moment of** O_2 $=$ **zero** D **)**

Q.7 Answer:

$25.5\ g$

Solution:

Strength $=$ **Normality** $\times$ **Eq. mass (eq. mass of** H_2O_2 **)** $= 25.5 gL^{-1}$

Q.8 Answer:

Solubility of polar compounds in water

Solution:

If structure of water is supposed to be linear instead of bent structure, then the dipole moment of water will become zero and the polar compounds will not be dissolved in it because "like dissolve like" i.e. polar compounds are soluble in polar solvents and non-polar compounds in non-polar solvents.

Q.9 Answer:

Alkali metals are less electropositive than alkaline earth metals.

Solution:

Because of small size, they are more closely packed in the crystal lattice. So they have comparatively stronger metallic bonds and so more denser and harder than alkali metals.

Q.10 Answer:

These metals can be isolated by the reduction of their oxides

Solution:

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Sodium and potassium can be prepared by the electrolysis of their fused chlorides but not from aqueous solutions of their salts, because these metals are very reactive.

Q.11 Answer:

Sodium oxide is more basic then magnesium oxide

Solution:

The thermal stability of calcium carbonate is more as compared to that of beryllium carbonate. The ionic potential (ϕ) value of Be^{2+} is more than that of Ca^{2+}. So Be^{2+} attracts the oxygen of $CO_3{}^{2-}$ more and on heating beryllium carbonate losses CO_2 more easily.

Q.12 Answer:

$8.4^{\circ}E$

Solution:

(i) $1^{\circ}E = 14.28ppm; ? = 120pm$

$\frac{120}{14.28} = 8.4^{\circ}E$

Q.13 Answer:

11.2

Solution:

(ii)

$CaCO_3 \quad CaO$

$100mg \quad 56mg$

$200mg \quad ?$

$\text{Amount of CaO per lit } = \frac{200 \times 56}{100} = 112mg$

$\text{But } 10mg \text{ of CaO per lit } = 1dGH, \quad 112mgCaO = ?$

$\frac{112}{10} = 11.2dGH$

Q.14 Answer:

178.6 ppm

Solution:

(iii) $10dGH = 100mgCaO$ **per lit. but** $56mgCaO = 100mgCaCO_3, 100mgCaO =$ **?**

$\frac{100 \times 100}{56} = 178.6ppm$

Q.15 Answer:

$CaCO_3$

Solution:

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$$\underset{(X)}{CaCO_3} \xrightarrow{\Delta} CaO + \underset{\text{Colourless gas}}{CO_2}$$

$$\underset{\text{Residue}}{CaO} + H_2O \longrightarrow \underset{(Y)}{Ca(OH)_2}$$

$$Ca(OH)_2 + \underset{\text{excess}}{2CO_2} \longrightarrow \underset{(Z)}{Ca(HCO_3)_2}$$

$$Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + CO_2 + H_2O$$

Q.16 Answer:

$Ca(OH)_2$

Solution:

$$\underset{(X)}{CaCO_3} \xrightarrow{\Delta} CaO + \underset{\text{Colourless gas}}{CO_2}$$

$$\underset{\text{Residue}}{CaO} + H_2O \longrightarrow \underset{(Y)}{Ca(OH)_2}$$

$$Ca(OH)_2 + \underset{\text{excess}}{2CO_2} \longrightarrow \underset{(Z)}{Ca(HCO_3)_2}$$

$$Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + CO_2 + H_2O$$

Q.17 Answer:

$Ca(HCO_3)_2$

Solution:

$$\underset{(X)}{CaCO_3} \xrightarrow{\Delta} CaO + \underset{\text{Colourless gas}}{CO_2}$$

$$\underset{\text{Residue}}{CaO} + H_2O \longrightarrow \underset{(Y)}{Ca(OH)_2}$$

$$Ca(OH)_2 + \underset{\text{excess}}{2CO_2} \longrightarrow \underset{(Z)}{Ca(HCO_3)_2}$$

$$Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + CO_2 + H_2O$$

Q.18 Answer:

A-q,s; B-r ; C- s ; D-p;

Solution:

((A) - qs, (B) - r, (C) - s, (D) - p)

Hot metallic magnesium and barium dissolved in liquid sodium can absorb nitrogen. Sodium potassium alloy which is a eutectic mixture consisting of 72.2% K and melting at $-12.3^{\circ}C$ is used as a coolant in

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heat exchangers. Reduction of magnesium halide with molten potassium produces magnesium. Liquid lithium has large liquid range of $454\ \mathrm{K} - 1615\ \mathrm{K}$.

Q.19 Answer:

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Solution:

$$\mathrm{LiNO_3} \xrightarrow{\Delta} \mathrm{Li_2O} + \mathrm{NO_2} + \mathrm{O_2}$$
$$\mathrm{NaNO_3} \xrightarrow{\Delta} \mathrm{NaNO_2} + \mathrm{O_2}$$

Q.20 Answer:

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Solution:

$\mathrm{x} = 3$ **(**$\mathrm{K}, \mathrm{Rb}, \mathrm{Cs}$ **)**
$\mathrm{y} = 2$ **(Mg, Be)**
$\mathrm{u} = 5$ **(Concentrated solution of all alkali metals in liquid ammonia is blue coloured solution)**

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